Jason Pan

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Education

University of Waterloo **Sep. 2025 – Present**

Candidate for Bachelor of Computer Science *Waterloo, Ontario*

- GPA: 3.86/4.00
- Relevant Courses: Designing Functional Programs (Racket), Algorithm Design and Data Abstraction (C), Object-Oriented Software Development (C++)

Technical Skills

Languages: Python, C, C++, HTML/CSS, JavaScript, SQL
Frameworks: React, React Native, Next.js, Flask, FastAPI, Tailwind CSS, PyTorch
Tools: Git, PostgreSQL, Supabase, Docker, Vite, Node.js, Railway, Figma

Experience

Full Stack Intern **May 2025 – August 2025**

Garde Robe Inc *New York, NY (Remote)*

- Rebuilt the web (Next.js) and mobile app (React Native/Expo) UI/UX from Figma design specs covering core flows and features across 25+ screens to prepare for a full relaunch replacing the company's live Bubble app.
- Designed a Shopify checkout extension and Cloudflare Workers backend with HMAC-verified webhooks and GDPR compliance handling to capture and match post-purchase orders into user closets via the Shopify Admin API.
- Developed a full in-app notification system and integrated an AI-powered auto-categorization pipeline for closet items.
- Benchmarked 5 third-party scraping services against the company's existing scraping architecture across 146 product URLs from 30 retailers with findings reflecting ∼15% higher accuracy and ∼27% lower latency.

Minecraft Game Development Coding Instructor **July 2025**

Alberta Junior AI Training Center *Edmonton, Alberta*

- Designed and delivered a 10-day introductory programming course with daily lessons and quizzes using a block-based coding editor that connects directly to a Minecraft Java Edition server.
- Taught fundamental CS concepts (functions, loops, variables, parameters, Cartesian/polar coordinate systems) through hands-on exercises building 3D structures ranging from towers and pyramids to amusement parks and pirate galleons.
- Directed a project for students to build their own Minecraft minigame, reinforcing iterative design and debugging skills.

Projects

Chess ML | *PyTorch, FastAPI, React, TypeScript, PostgreSQL, Supabase* **August 2026 – Present**

- Built a residual CNN policy network in PyTorch that predicts human moves conditioned on opening and Elo bracket.
- Engineered a data pipeline with python-chess that parses Lichess PGN dumps, tags each position with an opening and one of 6 per-player Elo buckets, and encodes boards as 8×8 tensors for training.
- Developed a FastAPI inference server and a React/TypeScript frontend where users practice 107 opening lines against a skill-matched model. Includes a guided move walkthrough, an in-browser Stockfish eval bar, and a games explorer.

LockNIn | *React, TypeScript, Flask, PostgreSQL, Gemini API* **January 2026 – Present**

- Deployed a full-stack study tracking platform used by 15+ students to log study sessions, manage homework and events, and track progress through analytics dashboards, powered by a Flask REST API with 50+ endpoints.
- Integrated the Gemini API with JSON-constrained prompting to parse course syllabi into structured deadlines and recurring events that users can review, edit, and import directly into their calendar.
- Built real-time messaging over WebSockets with Flask-SocketIO room-based broadcasting between friends.
- Automated email deadline reminders with APScheduler background jobs and thread locks to prevent duplicate sends.

Certifications

Introduction to Artificial Intelligence with Python | *Harvard CS50* **July 2026**

Google AI Professional Certificate | *Google* **May 2026**

Cybersecurity/Linux Essentials | *Cisco Networking Academy* **October 2023**